Review: Points, lines, and planes

1

a $\overleftrightarrow{FG}$ or $\overleftrightarrow{GF}$

b $\overrightarrow{FG}$

c $\overline{EG}$ or $\overline{GE}$

d $\overrightarrow{FE}$

e $\overleftrightarrow{EF}$ or $\overleftrightarrow{FE}$

f $\overline{EF}$ or $\overline{FE}$

2 Answers may vary:

a One such pair of points is C and K because they both lie on the same line.

b One such pair of points is I and J because they both lie on the same line.

3 $\overleftrightarrow{DC}$

4 $\overline{DF}$

5

a True

b True

c True

d False

6

a Parallel

b Not parallel

c Not parallel

d Not parallel

7

a True

b Not enough information

c True

d True

Additional questions

8 Answers may vary:

- **a** Points A, E, and H are collinear, as they all lie on the same line.
- **b** $\overleftrightarrow{AF}$ is one such line.
- **c** $\overrightarrow{AB}$ is one such ray.
- **d** Ray $\overrightarrow{BE}$ can also be called ray $\overrightarrow{BC}$.

9 Answers may vary:

- **a** $\overleftrightarrow{PQ}$ is contained in the plane
- **b** $\overleftrightarrow{AC}$ intersects the plane
- **c** Points P, Q, R, and S are all coplanar
- **d** Points A, Q, and C are collinear
- **e** $\angle PQR$ is contained in the plane
- **f** $\angle PQA$ is not contained in the plane
- **g** $\overrightarrow{QA}$ and $\overrightarrow{QC}$ are opposite rays

10 Plane XYZ and Plane R are two ways of naming this plane.

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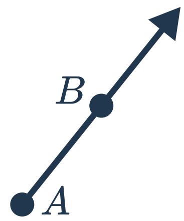
- **a** Point B
- **b** Point B
- **c** Line $\overleftrightarrow{XZ}$
- **d** Line $\overleftrightarrow{QR}$

12

a



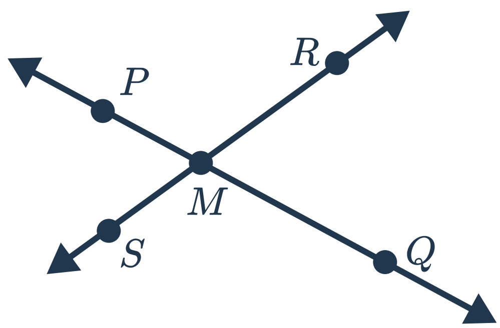
b $\overrightarrow{AB}$. The naming will change depending on the label used for the second point.

c Two points are needed to name a ray.

d Yes, the order of the points used to name the ray matters.
A ray has an endpoint, and then extends infinitely in the direction of the other point. If the points are used in the other order, the ray would extend in the other direction. For example, $\overrightarrow{BA}$ would start at point B and extend infinitely in the direction of point A.

13

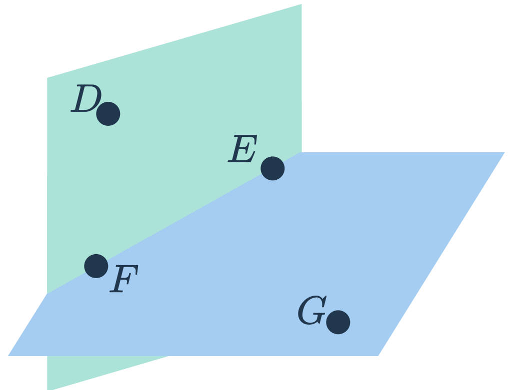
a



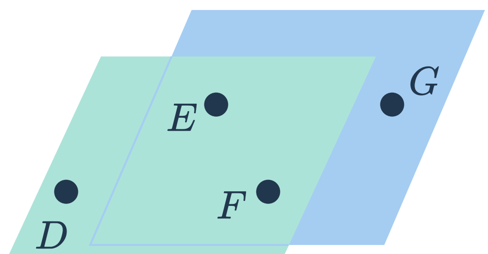
b $\overrightarrow{MP}$ and $\overrightarrow{MQ}$ are a pair of opposite rays. $\overrightarrow{MR}$ and $\overrightarrow{MS}$ are another pair of opposite rays.

14

a Here is a diagram of the two planes:

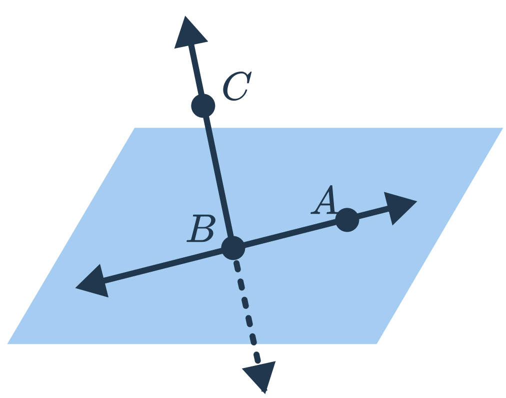


It is possible that point G also lies in the same plane as points D, E, and F, however. If this is the case, then the planes are the same:



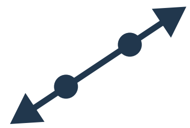
b For the situation in the first diagram, the intersection is the line $\overleftrightarrow{EF}$. For the situation in the second diagram, the intersection is the whole plane, which we could call plane DEF or plane EFG.

15

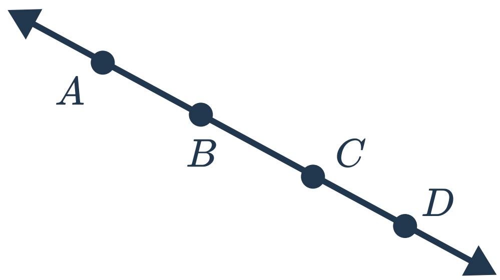


16

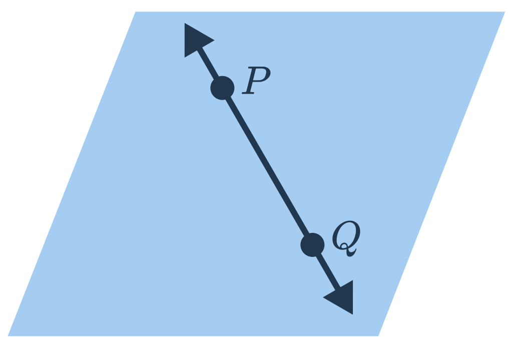
a True



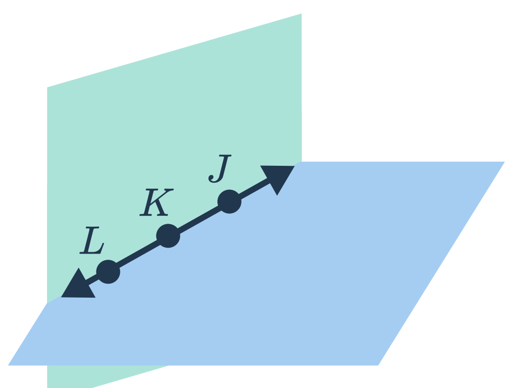
b False. Four distinct points can be collinear.



c True

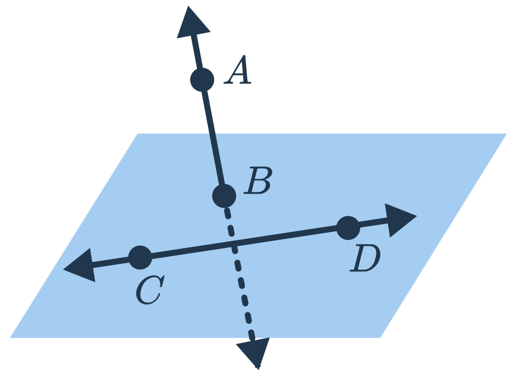


d False. Collinear points lie on a unique line, but not a unique plane.



e False. If three points are collinear, then they are contained in a single line, and there are many possible planes which contain them. The diagram in part (d) also serves as a counter-example for this statement.

f False. If the lines do not intersect, and are not parallel, then they can not be contained in a single plane.



17 Answers may vary:

a A laser pointer is a model for a ray. Light is emitted from the laster pointer, and then travels in a straight line off indefinitely, only stopping if it hits an object.

b The Golden Gate Bridge is a model for a line. Traffic can drive across it in both directions. It doesn't extend infinitely like a true straight line, however. Also, it curves very slightly due to the curvature of the Earth.

c A baseball pitch can be a model for a plane. The pitch has bounds, however, and doesn't extend infinitely in any direction like a true geometric plane.

d A highway passing under a bridge can be a model for a pair of skew lines. The highway and bridge are not parallel and, though they don't extend forever, if we could extend them infinitely they would never intersect.